



Set Theory & Fundamentals

Generated: August 23, 2026

SUBJECT

Mathematics Foundation T

SEMESTER

Semester 1

// Notes Summary

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{
  title: "Set Theory Fundamentals" semester: subject:
  "Mathematics Foundation to Computer Science I"
  subjectSlug: "mathfoundation1" unit: "Unit I: Set,
  Relation and Function" unitSlug:
  "unit1setrelationfunction" order: description:
  "Definition of sets, welldefined collections, roster
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NOTES THAT STICK.

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Topics Covered

Intermediate

Difficulty Level



Why this topic exists

Sets are the foundational building block of discrete mathematics and computer science. Core concepts in databases (relational algebra), query optimization, search algorithms, formal languages, and programming data structures (hash sets, dictionaries) borrow directly from set theory. Before defining relations, functions, or formal logic, a precise framework is required to define "a collection of objects." That framework is set theory.

What is a Set?

✦ [DEFINITION] Formal Set Definition

A **set** is a well-defined collection of distinct objects, considered as an entity in itself.

Two fundamental principles govern every set:

- **Well-defined:** There must be an objective criteria determining whether any given object belongs to the set.
- - *Invalid:* "The set of good programmers" (vague/subjective).
- - *Valid:* "The set of students with a GPA above 3.5" (objective).
- **Distinct:** Elements within a set are unique. Duplicate elements are ignored.

Examples: - $A = \{1, 2, 3, 4\}$ - $V = \{a, e, i, o, u\}$ (Set of English vowels) - $F = \{\text{Apple, Mango, Banana}\}$

In computational terms: A set represents a collection of unique data points with zero duplicate entries and explicit membership rules.

Set Representation

The standard method to represent a set is the **Roster (or Listing) Method**, where every element is listed explicitly inside curly braces $\{\}$, separated by commas.

Examples: - $A = \{2, 4, 6, 8\}$ - $N = \{1, 2, 3, 4, 5\}$ (First five natural numbers)



× [INVALID / PITFALL] Syntax Trap: Brackets vs Parentheses

Curly braces $\{\}$ strictly denote a set. Square brackets $[]$ and parentheses $()$ denote sequences, vectors, ranges, or ordered tuples in discrete mathematics and computer science.

(Note: Sets can also be expressed using **Set-Builder Notation**, which describes elements via a condition rather than an explicit list, e.g., $S = \{x \mid x \in \mathbb{N} \text{ and } x < 6\}$).

Element / Membership

An object contained within a set is called an **element** or **member** of that set.

- The symbol $\in$ denotes **membership** ("is an element of").
- The symbol $\notin$ denotes **non-membership** ("is not an element of").

Example: Given $A = \{10, 20, 30\}$: - $20 \in A$ (True: 20 belongs to A) - $50 \notin A$ (True: 50 does not belong to A)

Basic Types of Sets

1. Empty Set (Null Set) A set containing no elements. Denoted by $\{\}$ or $\emptyset$. -

Example: $A = \{x \mid x \text{ is an even prime number greater than } 2\} = \emptyset$

2. Singleton Set A set containing exactly one element. - *Example:* $A = \{5\}$

3. Finite Set A set whose cardinality (count of elements) is a fixed natural number.

The process of counting its elements terminates. - *Example:* $A = \{1, 2, 3, 4\}$ ($|A| = 4$)

4. Infinite Set A set whose elements cannot be completely listed or counted, as the collection continues indefinitely. - *Example:* $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$ (Set of all natural numbers)